Talk by Prof. Buckingham — Nov 8, '57

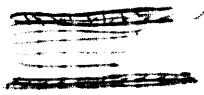
Speed $t \leq 10^{-8}$ sec — limits size

$n \sim 10^9$

Cost $\leq \$1 \times 10^{-3}$

Size: 10^{-6} Thickness — very low thermal noise.

$\sim 10^{-5}$

10^{-3}  insulator

3 pairs in — gas 10^4 bits per cc

Main limits

Range over 10^{12}

power-heat dissip.

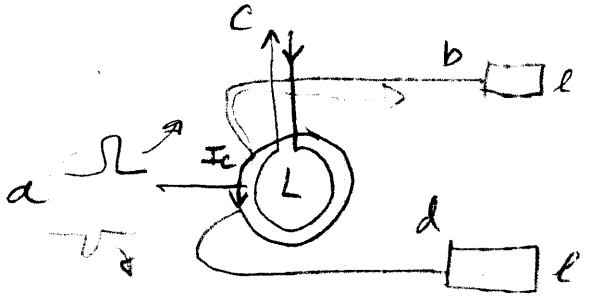
speed of conversion from super-normal

heat dissipated $\approx I_c \phi \omega$

↑ switching rate

voltage $\sim$ 1 volt

current $\sim 1 \times 10^{-2}$ amps



a P-Switch

a switch with mem.

current divide $1 - \frac{\ell}{L}$

Neg pulse goes upp, wrong { an "and" one wrong } different from switch. { an "or" either }

[need only 7 for binary adder.]

c	a	b	d
+	+	+	0
	−	0	−
−	+	0	+
	−	−	0

regular switch

b	d
+	0
−	0
0	+
0	−

Note: Cost of switches $\approx$ mem bits. circuits of diff logical type should be counted